
CS 545: Software Design Patterns

Course Code	CS 545
Course Name	Software Design Patterns
L-T-P-C	3-0-0-3
Prerequisites	UML and OO Programming
Intended for	UG & PG
Distribution	Elective for B.Tech. Electrical Engg.& Computer Sc. &Engg., MS/M.Tech., Ph.D.
Approval	11th Senate (Oct 2016)

Course Contents

- Review of UML and concepts of OO programming languages. [3 Lectures]
- Use Case and System Sequence Diagram. [2 Lectures]
- Design Methodology : Responsibility assignment. [3 Lectures]
- Design patterns classification and template. [3 Lectures]
- Creational Patterns. [6 Lectures]
- Structural Patterns. [6 Lectures]
- Behavioral Patterns. [6 Lectures]
- Architectural Patterns. [3 Lectures]
- Evaluative Patterns. [2 Lectures]
- Advance Topics such as Concurrency Patterns, Pattern Lang. [6 Lectures]

References

List is suggestive only, not mandatory

1. Craig Larman, **Applying UML and Patterns An Introduction to Object-Oriented Analysis and Design**, 3rd Edition.
2. Erich Gamma, Richard Helm, Ralph Johnson, John Vlissides, **Design Patterns Elements of Reusable Object-Oriented Software**, 2015(Paper back)
3. John Vlissides, **Pattern Hatching: Design Patterns Applied**
4. Martin Fowler, **Patterns of Enterprise Application Architecture**
5. **The Architecture of Open Source Applications**, Vol. 1 & 2